

Exercise 1: Ranking and Window Functions

Aim

To understand and implement SQL Window Functions such as ROW_NUMBER(), RANK(), DENSE_RANK(), OVER(), and PARTITION BY.

Problem Understanding

Window functions perform calculations across a set of rows without grouping the results. They are useful for ranking records, comparing values, and generating row numbers within different categories.

Window Functions Used

ROW_NUMBER()

Assigns a unique sequential number to each row within a category.

RANK()

Assigns the same rank to rows with equal values and skips the next rank.

DENSE_RANK()

Assigns the same rank to rows with equal values but does not skip any ranks.

Implementation

- Created a Products table.
- Inserted sample product records.
- Used PARTITION BY Category to rank products within each category.
- Used ORDER BY Price DESC to rank products from highest to lowest price.

SQL CODE:

```
CREATE TABLE Products (  
    ProductId INT PRIMARY KEY,  
    ProductName VARCHAR(50),  
    Category VARCHAR(50),  
    Price DECIMAL(10,2)  
);  
  
INSERT INTO Products VALUES  
(101, 'Laptop', 'Electronics', 65000),  
(102, 'Smartphone', 'Electronics', 45000),  
(103, 'Tablet', 'Electronics', 45000),
```

(104, 'Headphones', 'Electronics', 5000),
(201, 'Office Chair', 'Furniture', 12000),
(202, 'Study Table', 'Furniture', 18000),
(203, 'Bookshelf', 'Furniture', 18000),
(204, 'Sofa', 'Furniture', 35000),
(301, 'Rice', 'Groceries', 1200),
(302, 'Oil', 'Groceries', 1800),
(303, 'Sugar', 'Groceries', 1800),
(304, 'Tea Powder', 'Groceries', 950);

SELECT *

FROM (

SELECT

ProductId,

ProductName,

Category,

Price,

ROW_NUMBER() OVER (

PARTITION BY Category

ORDER BY Price DESC

) AS RowNumber

FROM Products

) RankedProducts

WHERE RowNumber <= 3;

SELECT

ProductId,

ProductName,

Category,

Price,

RANK() OVER (

PARTITION BY Category

```

ORDER BY Price DESC
) AS RankValue
FROM Products;

SELECT
    ProductId,
    ProductName,
    Category,
    Price,
    DENSE_RANK() OVER (
        PARTITION BY Category
        ORDER BY Price DESC
    ) AS DenseRankValue
FROM Products;

```

Comparison

Function	Handles Ties	Skips Rank
ROW_NUMBER()	No	No
RANK()	Yes	Yes
DENSE_RANK()	Yes	No

Output

```

Status : Successfully executed

Time:      Memory:
0.0000 secs  4.644 Mb

Your Output
101|Laptop|Electronics|65000|1
102|Smartphone|Electronics|45000|2
103|Tablet|Electronics|45000|3
204|Sofa|Furniture|35000|1
202|Study Table|Furniture|18000|2
203|Bookshelf|Furniture|18000|3
302|Oil|Groceries|1800|1
303|Sugar|Groceries|1800|2
301|Rice|Groceries|1200|3

```

```
101|Laptop|Electronics|65000|1
102|Smartphone|Electronics|45000|2
103|Tablet|Electronics|45000|2
104|Headphones|Electronics|5000|4
204|Sofa|Furniture|35000|1
202|Study Table|Furniture|18000|2
203|Bookshelf|Furniture|18000|2
201|Office Chair|Furniture|12000|4
302|Oil|Groceries|1800|1
303|Sugar|Groceries|1800|1
```

```
301|Rice|Groceries|1200|3
304|Tea Powder|Groceries|950|4
101|Laptop|Electronics|65000|1
102|Smartphone|Electronics|45000|2
103|Tablet|Electronics|45000|2
104|Headphones|Electronics|5000|3
204|Sofa|Furniture|35000|1
202|Study Table|Furniture|18000|2
203|Bookshelf|Furniture|18000|2
201|Office Chair|Furniture|12000|3
```

Conclusion

This exercise demonstrates how SQL Window Functions can be used to rank products within categories. These functions are useful for reporting, analytics, and retrieving top-performing records from a dataset.